ASRA

Across-Species Risk Assessment

When animal biology does not cleanly translate to humans.

ASRA is an evidence-grounded AI investigator for preclinical translation. It helps researchers examine where **animal evidence may not reliably transfer to humans** by combining four biological evidence layers, structured tool outputs and a multi-agent review process. ASRA **does not predict drug safety**; it identifies translational risk signals, evidence gaps and uncertainty.

01 Problem and who it helps

Problem

Traditional animal-to-human extrapolation can miss **differences in protein conservation, cellular localisation and gene expression**. These differences can weaken biological assumptions behind preclinical decisions.

Who it helps

Preclinical researchers, pharmacologists and drug-development teams who need an evidence-backed view of whether a target behaves similarly across species.

02 Data and evidence layers

1. Protein conservation

Ensembl + Biopython for orthologues, sequence identity and alignment coverage.

2. Subcellular localisation

UniProt for where the protein is reported to be located within the cell.

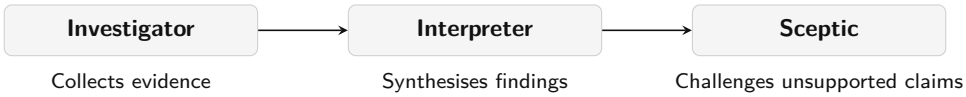
3. Cell-type expression

CZ CELLxGENE for which cell populations express the gene.

4. Tissue expression

GTEx + Bgee for tissue-level expression context and available cross-species evidence.

03 System design



Python computes biological measurements; Gemini agents coordinate evidence gathering, explanation and critique. Structured JSON keeps tool outputs auditable, and a revision loop allows the sceptic to challenge unsupported conclusions.

04 Implementation and evaluation

336	21	0
Backend tests passed	Tests skipped	Test failures

The prototype combines a **Python backend, Gemini agents and a web interface**. On the combined validation branch, backend tests passed with zero failures. Biological checks showed **79.39% identity** for **SLC29A1 human/mouse**, **98.18% identity** for **CD28 human/rhesus**, and preserved **CYP3A4/mouse ambiguity** rather than forcing an unsupported match.

05 Safeguards and limitations

Safeguards

Source identifiers and citations, explicit missing or ambiguous evidence, and sceptic review of claims. The system is designed to **show uncertainty rather than hide it**.

Limitations

ASRA **does not establish clinical safety or efficacy**. Sequence similarity does not prove functional equivalence. Findings require expert interpretation and experimental validation.